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Inventor(s)	Ponick; Matheus et al.

Knob mechanism for gas cooktop

Abstract

A locking mechanism to prevent inadvertent gas flow for a knob for a gas cooktop may include a stem configured to abut the knob in the resting position to prevent actuation of the knob, a locking mechanism movable between a locked position preventing translation of the stem to prevent against unintentional rotation of the knob, and a released position allowing translation of the stem and thus activation of the flow of gas from the burner, the locking mechanism including a rotatable mechanism wherein rotation of the rotatable mechanism moves the locking mechanism from the locked position to the released position, and a button arranged spaced and separate from the knob and attached to a rod extending downward from the button to engage the rotatable mechanism where the rod rotates the rotatable mechanism to translate the locking mechanism to the released position to allow the knob to be depressed in response to actuation of the button and the knob to prevent against unintentional rotation of the knob.

Inventors:	Ponick; Matheus (Joinville, BR), Schultz; Rodrigo (Joinville, BR)
Applicant:	WHIRLPOOL CORPORATION (Benton Harbor, MI)
Family ID:	87426726
Assignee:	Whirlpool Corporation (Benton Harbor, MI)
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Primary Examiner: McAllister; Steven B

Assistant Examiner: Carter; Amy E

Attorney, Agent or Firm: Brooks Kushman, P.C.

Background/Summary

TECHNICAL FIELD Described herein are locking systems for a knob mechanism for gas cooktops.

BACKGROUND

(1) A cooking appliance is used to cook meals and other foodstuffs on a cooktop or within an oven. The cooking appliance typically includes various control switches and electronics to control the heating elements of the cooking appliance.

SUMMARY

(2) A knob assembly for a gas cooktop may include a knob configured to control a flow of gas from a burner of a cooktop starting at a resting position, a stem configured to abut the knob in the resting position to prevent actuation of the knob, a locking mechanism having a locked position preventing translation of the stem and unintentional rotation of the knob and a released position allowing translation of the stem and thus activation of the flow of gas from the burner, the locking mechanism including a rotatable mechanism, wherein rotation of the rotatable mechanism moves the locking mechanism from the locked position to the released position, and a button arranged spaced and separate from the knob and attached to a rod extending downward from the button to engage the rotatable mechanism where the rod rotates the rotatable mechanism to translate the locking mechanism to the released position to allow the knob to be depressed in response to actuation of the button and the knob to prevent against unintentional rotation of the knob.

(3) In one embodiment, the locking mechanism is moved away from the stem and towards the rod in response to rotation of the rotatable mechanism by the rod.

(4) In another example, a spring is arranged at the stem to bias the knob in the resting position.

(5) In a further embodiment, the locking mechanism is a lever extending perpendicular to the stem.

(6) In one embodiment, the rotatable mechanism is a wheel arranged coplanar with the rod such that the wheel moves with actuation of the rod and pulls the locking mechanism away from the stem.

(7) In another example, the rotatable mechanism includes a torsion spring configured to bias the rotatable mechanism in the locked position.

(8) A locking mechanism to prevent inadvertent gas flow for a knob for a gas cooktop may include a stem configured to abut the knob in the resting position to prevent actuation of the knob, a locking mechanism movable between a locked position preventing translation of the stem to prevent against unintentional rotation of the knob, and a released position allowing translation of the stem and thus activation of the flow of gas from the burner, the locking mechanism including a rotatable mechanism wherein rotation of the rotatable mechanism moves the locking mechanism from the locked position to the released position, and a button arranged spaced and separate from the knob and attached to a rod extending downward from the button to engage the rotatable mechanism where the rod rotates the rotatable mechanism to translate the locking mechanism to the released position to allow the knob to be depressed in response to actuation of the button and the knob to prevent against unintentional rotation of the knob.

(9) In one embodiment, the locking mechanism is movable away from the stem and towards the rod in response to rotation of the rotatable mechanism by the rod.

(10) In another example, a spring is arranged at the rod to bias the knob in the resting position.

(11) In a further embodiment, the locking mechanism is a lever extending perpendicular to the stem.

(12) In one embodiment, the rotatable mechanism is a wheel arranged coplanar with the rod such that the wheel moves with actuation of the rod and pulls the locking mechanism away from the stem.

(13) In another example, the rotatable mechanism includes a torsion spring configured to bias the rotatable mechanism in the locked position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments of the present disclosure are pointed out with particularity in the appended claims. However, other features of the various embodiments will become more apparent and will be best understood by referring to the following detailed description in conjunction with the accompanying drawings in which:

(2) FIG. 1 illustrates an example cooktop, such as a gas cooktop;

(3) FIG. 2 illustrates an example knob assembly of the cooktop of FIG. 1;

(4) FIG. 3 illustrates a cross-sectional view of the knob assembly of FIG. 2 taken along A-A', where the locking mechanism is in a locked position;

(5) FIG. 4 illustrates a schematic diagram of a portion of the knob assembly of FIG. 2, where the locking mechanism is in a locked position;

(6) FIG. 5 illustrates a schematic diagram of a portion of the knob assembly of FIG. 2, where the locking mechanism is in a released position; and

(7) FIG. 6 illustrates a cross-sectional view of the knob assembly of FIG. 2, where the locking mechanism is in a released position.

DETAILED DESCRIPTION

(8) As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

(9) Knobs in gas cooktops and freestanding ranges are often sensitive to actuation and may be accidentally turned on. While not intended, such accidental actuations or rotations may release gas unknowingly. As described in detail herein, an improved system allows for an additional locking mechanism to prevent the unintended actuation during of the knob. This locking mechanism may ensure that any release of gas is in response to actuations that are deliberate and not accidental.

(10) Current knobs may have a two-step initial activation, including pushing the knob and then rotating the knob to a desired position to release gas. The knob, and consequently the gas valve, could be initially activated by accident, allowing for gas leakage.

(11) Disclosed herein is a knob assembly having a locking assembly that requires an actuation of a third mechanism, in addition to the two-step initial activation, for gas release. In this example, the third mechanism may include a button arranged separate and distinct from the knob assembly. The knob assembly may also include a stem, cooperating with the button, that locks the knob by preventing the knob from being pushed in the first place. Actuation of the button may release the stem from engaging the knob to allow the knob to be depressed and subsequently rotated to activate the gas flow.

(12) In practice, a rod or other suitable mechanical feature may block the stem from moving downward when the knob is locked. The rod may be released by pressing of the button, in some examples, by rotation of the rod body caused by pressing of the button. The rod may have a torsion spring attached to it, so the rod may be biased to the locked position when the knob is at the off or resting position. However, the knob may return to the locked position freely after it was released and turned to the desired location. In this case, the stem may automatically return to the locked position with the knob. The return of the knob to the locked position may occur automatically using a spring mechanism engaging the stem and/or rod, not requiring another step to turn the knob off.

(13) Thus, if the activation was accidental, the rod may block the stem from moving downward and thus prevent depression of the knob. This prevents the knob from leaving its initial locked position and releasing gas without the spark.

(14) FIG. 1 illustrates an example cooktop **100**, such as a gas range assembly. The cooktop **100** may include a cooking area **104** having a plurality of burners **106**, each controlled by a knob assembly **101** having a knob **102**. Each separately controlled burner **106** is dedicated to supplying heat to a corresponding area of the cooking area **104**. The heat supplied to each separately controlled heating area is controlled such that a command to change the heat supplied to it may not change the amount of heat supplied to any other separately controlled cooking area **104**. In the example of FIG. 1, the cooktop **100** has four separately controlled cooking areas **104**, but more or fewer cooking areas **104** may be included.

(15) One or more grates **110** may be arranged above the cooking area **104** in order to maintain cookware thereon a predefined distance above the burners **106**. Each grate **110** may be made of metal, iron, or some other thermally conductive element. Each burner **106** may be operable to heat to desired cooking temperatures. In an example, each knob **102** is configured to control the flow of gas to a respective one of the burners **106**. The knobs **102** may be labeled to allow a user to identify which knob **102** controls which of the burners **106**. The burners **106** are configured to generate controlled flames that may be used to heat cookware arranged on the grate **110**. The magnitude of the flame generated by the burners **106** is proportionate to the amount of gas flowing to the burners **106**. A user may adjust the flow of gas to the burners **106** using the knobs **102**. As the user rotates each of the knobs **102**, a gas control valve (not shown) changes the amount of gas flowing to the corresponding burner **106**.

(16) While the knobs **102** in the example of FIG. 1 are illustrated as being arranged on top of the cooking area **104**, the knobs may also be arranged on a front surface of a manifold of the cooktop **100**. The knobs **102** may include markings therearound to indicate a certain level of heat being supplied by the burner **106** relative to the rotational position of the knob **102**. For example, markings associated with a high, medium, low, simmer settings may be included. Each knob **102** has a face **112** with a grip **114** extending outwardly from the face **112**. It should be appreciated that in other embodiments each knob **102** may be contemplated, such as the knob **102** being shaped as a cylinder or oval without a grip. Although not shown, a grate cover may be arranged over the grates to create a surface that protects the cooking area **104** as well as providing for additional surface space.

(17) FIG. 2 illustrates an example knob assembly **101** of the cooktop of FIG. 1. The knob assembly **101** may include the knob and a button **107**, or other suitable actuation element. The button **107** may be arranged separate but near the knob **102** so that actuation of the button **107** and the knob **102** can be achieved concurrently or near concurrently (e.g., the button **107** being engaged prior to engagement of the knob **102**). Upon concurrent or near concurrent depression of the knob **102** and the button **107**, a locking mechanism (not shown in FIG. 2) may move from a locked state or position to a released position or state and allow the knob **102** to rotate to the desired position to release gas to the burner **106**. In the locked state, the knob **102** may be prevented by the locking mechanism from being depressed and thus prevented from rotating. Because of this, inadvertent actuation of the knob **102** may be prevented.

(18) FIG. 3 illustrates a cross-sectional view of the knob assembly **101** of FIG. 2, where the knob assembly **101** includes the locking mechanism in the locked state. As explained, the knob **102** may include grip **114** configured to engage with the user's fingers to apply pressure to rotate and actuate the knob **102**. The grip **114** may also indicate a rotational location of the knob **102**, thus indicate the flow level of gas such as high, medium, low, etc. When a user applies pressure to the knob **102**, the knob **102** may be pushed downward into a bezel **138** and subsequently rotated.

(19) The knob **102** may define a hollow interior. A support cylinder **105** may extend from the underside of the grip **114** through the inside center of the knob **102**. The cylinder **105** may form a hollow opening having a generally cylindrical shape and a flat side. The cylinder **105** may be configured to receive a post **109** during assembly of the knob assembly **101** onto the cooktop **100**. Although not shown, a spring may be arranged within the cylinder **105** to bias the knob **102** away

from the cooktop **100** in the resting position (e.g., not pushed into the bezel **138**).

(20) A stem **130** may extend from within the cooktop **100** through to the surface and be configured to abut the underside of the knob **102**. In one example, the hollow interior of the knob **102** may define a stop **136** configured to align with the stem **130** and abut the stem **130** in the resting and locked position. The stem **130** may alternatively extend from the knob **102** and into the cooktop. The stem **130** is arranged generally perpendicular with the cooktop surface. Within the cooktop **100**, a spring **132** may be arranged at a distal end of the stem **130** to bias the stem **130** such that the knob is biased in the resting position. When compressed, the spring **132** may create some resistance when the user depresses the knob **102**, but still allow the user to depress and subsequently rotate the knob **102**.

(21) FIG. 4 illustrates a schematic diagram of a locking assembly **140** of the knob assembly **101** of FIG. 2, where the locking assembly **140** is in a locked state. The locking assembly **140** may be arranged within the cooktop and may be used to prevent actuation of the knob **102** without dual actuation of the button **107** and the knob **102**. The locking assembly **140** may include the stem **130** and spring **132**, as well as the button **107**. Further, the locking assembly **140** may include a locking mechanism **142**, a rod **144**, and a rotatable mechanism **146**.

(22) The rod **144** may operatively extend from the button **107** and engage with the rotatable mechanism **146**. The rotatable mechanism **146** may be fixed at a pivot and rotatable about that pivot (not shown). In one example, the rotatable mechanism **146** may be a wheel, or semicircular shape. In other examples the rotatable mechanism **146** may be other shapes. The rotatable mechanism **146** may be generally flat or planar and extend along and coplanar with the rod **144**.

(23) Both the rod **144** and the locking mechanism **142** may be fixed to the rotatable mechanism **146**. In the locked position or locked state, the locking mechanism **142** abuts the distal end of the stem **130**. This prevents the stem **130** from translating towards the spring or further into the cooktop **100** and thus prevents depression of the knob **102**. The spring **132** also biases the locking mechanism against the stem **130** in the locked position. The stem **130** may be a solid cylinder or block, and also may be a hollow tub-like support arranged on a post **148**. In this example, the stem **130** may selectively move along the post **148**.

(24) The locking mechanism **142** may be a lever extending perpendicular to the stem **130**, and in some examples may extend perpendicular to the rod **144**. The locking mechanism **142** and rod **144** may be made of rigid materials such as metal, plastic, resin, etc., and may be configured to hold their shape in the heated environment of the cooktop. In one example, the locking mechanism **142** is a metal rod or post. In another example, the lever is a metal bracket.

(25) The locking mechanism **142** may be maintained on a support structure **134** to aid in maintaining the locking mechanism **142** perpendicular to the stem **130** and/or the rod **144**. The support structure **134** may not be fixed to the locking mechanism **142** but instead simply support the locking mechanism and allow the locking mechanism **142** to move across the support structure between the locked and released positions.

(26) FIG. 5 illustrates a schematic diagram of the locking assembly **140** of the knob assembly **101** of FIG. 2, but where locking assembly **140** is in a released state. In this example, the locking mechanism **142** may be in the released state where the locking mechanism **142** is released from the distal end of the stem **130** such that the stem **130** may be translated into the cooktop when the knob **102** is depressed. Because the locking mechanism **142** is pulled toward the rod **144** and away from the stem **130** in the released state (via actuation of the button **107**), the locking mechanism **142** no longer abuts the stem **130** and allows the stem **130** to depress the spring **132**. That is, translation of the rod **144** caused by actuation of the button **107** rotates the rotatable mechanism **146**. This in turn pulls the locking mechanism **142** away from the stem **130** and allows for depression of the knob **102** (via compression of the spring **132**).

(27) Once the locking mechanism is in the released position or state, the knob **102** may be rotated and move freely to its desired position to allow the desired flow of gas. Once the knob **102** is

initially depressed and rotated, the knob **102** may be further rotated without depression of the button **107**. That is, each time the user wishes to adjust the flow of gas, it is not necessary to again depress the button **107**. Further, the knob **102** may be returned to the resting or closed position without actuation at the button. As such, actuation of the button **107** may only be required when actuating the knob **102** from the resting position where the gas flow is off. Including to rotate the knob back to a closed position of the knob will turn the gas flow off.

(28) As illustrated in FIG. 5, actuation of the button **107** forces translation of the rod **144**. The rotatable mechanism **146** is forced to rotate about the pivot point by the rod **144** and concurrently the rotatable mechanism **146** pulls the locking mechanism **142** from its blocking position at the stem **130** to an unblocked position. This unblocked position then allows the user to rotate the knob **102**, while also preventing against inadvertent actuation at the knob that may lead to a gas release.

(29) FIG. 6 illustrates a cross-sectional view of the knob assembly **101** of FIG. 2, where the locking assembly **140** is in a released state. As explained, the knob **102** may include the grip **114** configured to engage with the user's fingers to apply pressure to rotate and actuate the knob **102**. When a user applies pressure to the knob **102**, the knob **102** may be pushed downward into a bezel **138** and subsequently rotated.

(30) The knob **102** may define a hollow interior and the support cylinder **105** may extend from the underside of the grip **114** through the inside center of the knob **102**. The cylinder **105** may be configured to receive the post **109** during assembly of the knob assembly **101** onto the cooktop **100**. The stem **130** may extend from the cooktop **100** where the stem **130** is generally perpendicular with the cooktop surface. In this released position, the stem **130** may be depressed with the knob **102** because the locking mechanism **142** is not blocking the stem **130**. As explained above with respect to FIG. 4, the stem **130** may be a hollow tube arranged on and movable along the post **148**. The stem **130** may also be a solid cylinder or block. Within the cooktop **100**, the spring **132** may be arranged at a distal end of the stem **130** to bias the stem **130** in the resting position. When compressed, the spring **132** may create some resistance when the user depresses the knob **102**, but still allow the user to depress and subsequently rotate the knob **102**. The spring **132** may bias the return to the locking position, once the knob is rotated to the closed state.

(31) As illustrated in FIG. 6, the spring **132** is depressed with the knob since the locking assembly **140** is in the released state, allowing the user to freely actuate the knob **102**.

(32) In the resting position (i.e., where the knob **102** is not depressed, the button **107** is not actuated, and the locking assembly **140** is in the locked state), should the knob **102** be turned, bumped, etc., the locking assembly **140** will prevent the knob **102** from moving. Gas will not be released, and the bias may ensure that any depression and rotation of the knob are deliberate and release of gas is in response to actuations are not accidental. If the rotation was done on purpose by a user, the user must actuate the button **107**, and depress and continue to rotate the knob **102** until the spark starts to produce a sound and gas is flowing to the burner. After the initial rotation, the knob **102** may move freely to allow the user to select the desired position of the knob **102** for the desired gas flow, without additional actuation of the button **107**, as the locking mechanism is in the released state.

(33) Accordingly, a knob assembly is disclosed that prevents inadvertent actuation from releasing gas accidentally.

(34) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

(35) For purposes of description herein the terms "upper," "lower," "right," "left," "rear," "front," "vertical," "horizontal," and derivatives thereof shall relate to the device as oriented in FIG. 1. However, it is to be understood that the device may assume various alternative orientations and

step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(36) The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.

(37) Aspects of the present embodiments may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

Claims

1. A knob assembly for a gas cooktop, comprising: a knob configured to control a flow of gas from a burner of a cooktop starting at a resting position; a stem configured to abut the knob in the resting position to prevent actuation of the knob; a locking mechanism having a locked position preventing translation of the stem and unintentional rotation of the knob and a released position allowing translation of the stem and thus activation of the flow of gas from the burner, the locking mechanism cooperating with a rotatable mechanism, wherein rotation of the rotatable mechanism translates the locking mechanism from the locked position to the released position; and a button arranged spaced and separate from the knob and attached to a rod extending downward from the button to engage the rotatable mechanism, where the rod rotates the rotatable mechanism to move the locking mechanism to the released position to allow the knob to be depressed in response to actuation of the button and the knob to prevent against unintentional rotation of the knob, where the locking mechanism is moved laterally away from the stem and towards the rod in response to rotation of the rotatable mechanism by the rod.
2. The assembly of claim 1, further comprising a spring arranged at the stem to bias the knob in the resting position.
3. The assembly of claim 1, wherein the locking mechanism is a lever extending perpendicular to the stem.
4. The assembly of claim 1, wherein the rotatable mechanism is a wheel arranged coplanar with the rod such that the wheel moves with actuation of the rod and pulls the locking mechanism away from the stem.
5. The assembly of claim 1, wherein the locking mechanism creates a stop and blocks the stem from translation from the locked position.
6. A knob assembly to prevent inadvertent gas flow for a knob for a gas cooktop, comprising: a knob; a stem configured to abut the knob in a resting position to prevent actuation of the knob; a locking mechanism movable between a locked position preventing translation of the stem to prevent against unintentional rotation of the knob, and a released position allowing translation of the stem and thus activation of the flow of gas from the burner, the locking mechanism cooperating with a rotatable mechanism wherein rotation of the rotatable mechanism translates the locking mechanism from the locked position to the released position; and a button arranged spaced and separate from the knob and attached to a rod extending from the button to engage the rotatable mechanism, where the rod rotates the rotatable mechanism to move the locking mechanism to the

released position to allow the knob to be depressed in response to actuation of the button and the knob to prevent against unintentional rotation of the knob, where the locking mechanism is laterally movable away from the stem and towards the rod in response to rotation of the rotatable mechanism by the rod.

7. The knob assembly of claim 6, further comprising a spring arranged at the rod to bias the stem to the resting position.

8. The knob assembly of claim 6, wherein the locking mechanism is a lever extending perpendicular to the stem.

9. The knob assembly of claim 6, wherein the rotatable mechanism is a wheel arranged coplanar with the rod such that the wheel moves with actuation of the rod and pulls the locking mechanism away from the stem.

10. The knob assembly of claim 6, wherein the locking mechanism creates a stop and blocks the stem from translation from the locked position.
